

MONGO-DB lab3

Q) Create a database `universitydb` with collections:

- `students`
- `courses`
- `enrollment`
- `faculty`
- `attendance`

a)

Design a **students** collection where each document includes:

- `studentId`
- `name`
- `department`
- `semester`
- `skills`

Insert **at least 5 students** using `insertMany()`.

b)

Insert a student such that **if semester $\geq$ 5**, include an embedded document `internship` containing:

- `company`
- `duration`

c)

Retrieve **only**:

- name
- department
- skills
(Exclude `_id`)

d)

Find students who:

- Have **both Python and MongoDB** in `skills`
- Belong to **CSE department**

e)

Retrieve students who have **any subject** with:

- Grade = "A+"
- Credits ≥ 4

Database:

c)

```
university2> db.students.find({}, { name: 1, department: 1, skills: 1, _id:
0 }).pretty()
[
  { name: 'Archit', skills: [ 'C', 'Java' ] },
  { name: 'yash', skills: [ 'Python', 'MongoDB' ] },
  { name: 'Karan', skills: [ 'Python', 'C++' ] },
  { name: 'Neha', skills: [ 'AutoCAD' ] },
  { name: 'Arjun', skills: [ 'Python', 'MongoDB', 'NodeJS' ] },
  { name: 'utsav', skills: [ 'Python', 'MongoDB' ] }
]
```

d)

```
university2> db.students.find({ dept: "CSE", skills: { $all: ["Python", "MongoDB"] } }).pretty()

university2> db.students.find({ dept: "CSED", skills: { $all: ["Python", "MongoDB"] } }).pretty()
[
  {
    _id: ObjectId('69807ce40866f020dd628ca1'),
    studentId: 102,
    name: 'yash',
    dept: 'CSED',
    sem: 5,
    skills: [ 'Python', 'MongoDB' ],
    subjects: [
      { name: 'DBMS', grade: 'A+', credits: 4 },
      { name: 'OS', grade: 'A', credits: 3 }
    ]
  },
  {
    _id: ObjectId('69807ce40866f020dd628ca4'),
    studentId: 105,
    name: 'Arjun',
    dept: 'CSED',
    sem: 7,
    skills: [ 'Python', 'MongoDB', 'NodeJS' ]
  },
  {
    _id: ObjectId('69807d270866f020dd628ca5'),
    studentId: 106,
    name: 'utsav',
    dept: 'CSED',
    sem: 5,
    skills: [ 'Python', 'MongoDB' ],
    internship: { company: 'Infosys', duration: '6 months' }
  }
]
```

e)

```
university2> db.students.find(
...     {
...         subjects: {
...             $elemMatch: {
...                 grade: "A+",
...                 credits: { $gte: 4 }
...             }
...         }
...     }
... ).pretty()
...
[
  {
    _id: ObjectId('69807ce40866f020dd628ca1'),
    studentId: 102,
    name: 'yash',
    dept: 'CSED',
    sem: 5,
    skills: [ 'Python', 'MongoDB' ],
    subjects: [
      { name: 'DBMS', grade: 'A+', credits: 4 },
      { name: 'OS', grade: 'A', credits: 3 }
    ]
  }
]
```